The XELOR agent system

Nine agents, nineteen tools and seven missions — and the machinery that keeps all of it inside the authority of the person who asked.

Covers: what XELOR is, the nine agents, the architecture, every capability, all seven mission graphs, how a run actually executes, the trust model, and what is honestly not built.
Companion set: one guide per agent.
Truth standard: the repository as it stands on 8 August 2026.

MASTER

01 · THE IDEA

What XELOR is

A manufacturing system of record, with a controlled layer that reads across it

The plain version

A factory's information sits in separate places — orders, stock, plans, the shop floor, quality, machines, money, people. Each has its own rules and each is right about its own subject. What nobody has is one honest answer to a question that crosses all of them. XELOR's agents exist for that question, and for nothing else: they do not replace the modules, they read them under the asker's own permissions and put the pieces side by side.

9 named agents

22 product modules

19 registered tools

7 mission graphs

Business records → **Module rules** → **Registered tools** → **Specialists** → **ONYX joins** → **Human decides**

The modules stay in charge

Sales owns orders, Inventory owns stock, Accounts owns journals. An agent calls those services; it never gets its own copy of the data or its own way in.

The agents make the crossing visible

Each contributes only what its own area can prove. ONYX joins it, HEXA checks it, and a person decides anything with a consequence.

Nine agents, one system

Each has a full companion guide

ONYX · Supervisor

Turns a business goal into a bounded plan, assigns the right specialists, joins their evidence and presents one understandable result.

HEXA · Governance

Holds identity, tenant isolation, permissions, approvals and the tamper-evident record, so every mission stays inside the authority of the person who started it.

MICA · Sales & Product Care

Explains what a customer ordered, what was promised, what has shipped, and which manufactured-product warranty or after-sales obligation remains open.

SPAR · Supply

Explains what stock exists, what has been ordered from suppliers, and where a shortage threatens a plan or a promise.

AXLE · Planning

Connects what the product is made of with what demand requires, then says what to make, what to buy, by when — and which dates cannot be met.

KILN · Operations & Quality

Explains what the plant is making, whether it passes inspection, and whether the machines will still be there tomorrow.

RASP · Finance & People

Explains the accounting truth, the cash position, what is already committed, and the payroll consequence behind an operating decision.

RELAY · Managed Services

Coordinates the service around XELOR—from design and onboarding to incidents, changes, service reviews and improvement—while the accountable specialist still performs the technical work.

ACHILES · Platform Assurance

Privately checks whether XELOR is responding, preserves the result history and hands failed evidence to the people and agents responsible for incident coordination and repair.

The counterintuitive part, and the best proof the design is real

ONYX is the supervisor, and it is the LEAST powerful agent in the system. It holds one read tool and is not on the allow-list for the one capability that can cause an effect. It can convene a mission and write the summary; it cannot touch the business. A design where the coordinator accumulates power is the failure everyone expects from agent systems — this one gives it the smallest surface of all.

How authority is distributed

One supervisor

ONYX is the only agent that may delegate. The eight specialists cannot recruit each other or widen their own role.

One control layer

HEXA verifies before and after the human gate, and applies the same tenant, permission and approval rules to everybody.

Twenty-two modules under nine agents

The master gives the map; the agent guides explain each module in depth

Agent	Modules	What the group contributes
ONYX Supervisor	ONYX Copilot · Agent OS · AI Operations	Copilot handles one bounded question; Agent OS handles a cross-functional mission; AI Operations governs any model-assisted feature. They share identity, permissions, logging and refusal rules, but they do not share a hidden general-purpose action channel.
HEXA Governance	Organisation · Administration · Integration	Organisation supplies the legal and numbering context, Administration decides who may do what and proves the trail, and Integration carries external failures as explicit state. Together they are the control plane used by every business module and every agent run.
MICA Sales & Product Care	Sales · Customer Care & Warranty	Sales records what was promised and fulfilled; Customer Care & Warranty records what happened to the manufactured product after delivery. MICA links the same customer, order, installed base and quality evidence. RELAY separately owns operational incidents with XELOR itself.
SPAR Supply	Purchase · Inventory	Purchase creates an approved supply commitment; Inventory records the physical consequence. A receipt is useful only when both succeed in one transaction, and downstream modules move stock through Inventory rather than maintaining their own balances.
AXLE Planning	Engineering · Planning	Engineering defines what a product is; Planning applies demand, stock, supply, calendars and policy to decide what must happen by date. Production receives proposals and a pinned structure, so later master-data changes cannot rewrite work already released.
KILN Operations & Quality	Production · QMS & Audit · Maintenance	Production records the work, Quality decides whether the result is acceptable, and Maintenance explains whether the equipment can sustain the plan. Inventory movements are shared evidence, while quality and downtime conclusions remain deterministic and human-accountable.
RASP Finance & People	Accounts · Employee Spend · People · Working Capital	Accounts is the posted truth; Employee Spend and People create controlled obligations and postings; Working Capital turns the available evidence into reviewable priorities and scenarios. RASP must label the difference between a ledger fact, an assumption and an illustrative demo value.
RELAY Managed Services	Managed Services	RELAY owns coordination, elapsed time and customer communication. HEXA, ONYX, MICA, SPAR, AXLE, KILN or RASP remains the technical/business owner for its domain and must supply closure evidence. A human owns contracts, credits and material promises.
ACHILES Platform Assurance	Private Platform Assurance	ACHILES detects and records only. RELAY owns incident severity, clock and customer updates. HEXA, ONYX or the affected specialist owns diagnosis and repair. A person keeps control of high-impact decisions.

One ownership rule prevents a great deal of confusion

The module that owns a record is the only place allowed to change it. A mission can combine Sales, stock, plan, quality, finance and service evidence, but it does not create a second cross-functional copy. The agent layer coordinates decisions; domain services remain the system of record.

How the modules complete one factory story

Control and evidence travel with the transaction

MICA	→	AXLE	→	SPAR	→	KILN	→	MICA / RASP	→	RELAY
customer promise		product + plan		supply + stock		make + check		deliver + account		assure service

1 · Commit

Sales stores the dated customer promise, GST result and credit decision. Planning receives demand; it does not reinterpret the order.

2 · Plan and supply

Engineering defines the BOM. Planning nets demand against stock and supply. Purchase approves a commitment; Inventory records the physical receipt.

3 · Execute and prove

Production pins components and operations. Quality snapshots the limits it judged. Maintenance preserves the uptime evidence behind delivery risk.

4 · Deliver and account

Dispatch posts stock-out and the invoice together. Accounts holds the receivable and later receipt; MICA's Customer Care & Warranty keeps the manufactured-product obligation after delivery.

HEXA surrounds every step

Verified identity, tenant fence, permission, approval and audit apply before the business service is allowed to touch a record.

ONYX crosses the loop safely

It asks the eight specialists for evidence, joins their answers and waits for the person. It does not take ownership away from any module.

RELAY keeps the service whole

It owns onboarding, incident clocks, hand-offs, customer updates, the change calendar and reviews; the affected specialist still owns the technical fix.

ACHILES watches quietly

It checks platform availability, records private evidence and stops. RELAY coordinates any incident; the technical owner diagnoses and repairs.

The Northstar proof

The seeded evidence follows one 120-unit PX-400 customer order through MRP, purchase, three production orders, a rejected inspection, 12 quarantined units, 28 dispatched units, an invoice, a receipt, a manufactured-product case, employee spend and a Decision Commander approval. The same identifiers reconcile across the modules.

How a question reaches trusted data

Every layer narrows what is possible and keeps the evidence

Person / signal → Mission → Specialist → Registry → Domain service → Tenant data → Audit

Experience	Copilot, dashboards, Decision Commander
Coordination	Graph catalogue, runtime, waves, joins, checkpoints
Governance	Identity, tenant fence, permissions, approval, kill switch, audit
Capability	A closed registry of 19 operations
Business	Sales, stock, planning, production, quality, maintenance, finance, people
Managed service	Catalogue, transition, incident/change coordination, customer communication, improvement
Platform assurance	Private deterministic checks, freshness and append-only availability history
Data	Tenant-fenced PostgreSQL, append-only ledgers, durable run records

There is no database agent

An agent cannot write a query. It names a registered capability; the runtime checks who is asking and whether they may; the capability calls a service whose rules are already in force. The reason the Copilot can refuse “ship the held units anyway” is not good judgement — it is that no endpoint exists that would take such an instruction.

All 19 capabilities

The complete registry — nothing else is callable

Capability	Mode	Who may call	What it returns	Permission
general.companies.read	Read	HEXA, ONYX	Company masters	general.company.read
sales.orders.read	Read	MICA	Sales orders	sales.order.read
inventory.on-hand.read	Read	SPAR	Stock balances	inventory.stock.read
planning.planned-orders.read	Read	AXLE	Planned orders	planning.mrp.read
production.orders.read	Read	KILN	Production orders	production.order.read
production.factory-connect.read	Read	KILN	Robot, AMR, gateway and material-dwell evidence	factory.connect.read
quality.inspections.read	Read	KILN	Inspections	quality.inspection.read
accounts.vouchers.read	Read	RASP	Journal vouchers	accounts.ledger.read
finance.cash-position.read	Read	RASP	Trial balance	accounts.ledger.read
quality.evidence.collect	Analyse	KILN	Evidence view	quality.inspection.read
finance.collections.prioritise	Analyse	RASP	Review queue	accounts.ledger.read
finance.forecast.simulate	Simulate	RASP	13-week scenario	accounts.ledger.read
quality.audit-plan.draft	Draft	KILN	Audit checklist	quality.inspection.read
quality.capa-plan.draft	Draft	KILN	Investigation draft	quality.inspection.read
quality.audit-pack.draft	Draft	KILN	Evidence manifest	quality.inspection.read
finance.funding-pack.draft	Draft	RASP	Draft manifest	accounts.ledger.read
managed-services.service-assurance.read	Read	RELAY	Service assurance view	managed_services.overview.read
platform-health.status.read	Read	ACHILES	Private platform status and history	platform_health.overview.read
agent.action.dispatch	**Execute**	HEXA, MICA, SPAR, AXLE, KILN, RASP, RELAY	Governed work item	agentos.run.operate

18 of 19 cannot change anything

Exactly one capability has a side effect, and a test asserts that count so it cannot drift. That one requires an approved human gate above it in the graph, and the engine checks the ancestry itself rather than trusting the capability to behave. Note also that ONYX and ACHILES are absent from the execute row.

Every registered graph

102 nodes in total, all written down before any run starts

Factory flow recovery

`factory.flow-recovery · 18 nodes · step budget 24 · timeout 300s · 0 dispatch nodes`

KILN reads stored robot and dwell evidence while MICA, SPAR, AXLE and RASP read business consequences; specialists assess; HEXA checks the declared boundary; a production supervisor approves the proposed simulator request; ONYX publishes a brief. The graph has no dispatch node and never contacts a controller.

Cross-functional readiness review

`foundation.cross-functional-readiness · 9 nodes · step budget 14 · timeout 300s · 0 dispatch nodes`

ONYX frames it · MICA and SPAR read in parallel · both assess · join · HEXA verifies · human approves · ONYX synthesises.

Nine-agent operating review

`operations.full-command-review · 21 nodes · step budget 29 · timeout 300s · 0 dispatch nodes`

All eight specialists read their own evidence at once, each assesses, ONYX joins, HEXA verifies four checks, a human approves, ONYX issues the brief.

Nine-agent controlled action mission

`operations.controlled-action-mission · 31 nodes · step budget 39 · timeout 600s · 7 dispatch nodes`

The read-propose-verify-approve-dispatch-verify contract. Eight reads and recommendations, one plan, a preflight, ONE human gate, then six domain work items plus one RELAY service-coordination item and an outcome check. ACHILES remains read-only.

Working Capital Review

`finance.working-capital-review · 9 nodes · step budget 18 · timeout 300s · 0 dispatch nodes`

RASP reads the trial balance, MICA the commitments, SPAR the stock holding; RASP simulates 13 weeks; HEXA verifies; a human approves the brief.

QMS & Audit Readiness

`quality.qms-audit-readiness · 8 nodes · step budget 18 · timeout 300s · 0 dispatch nodes`

KILN reads inspections, collects evidence and drafts a pack; HEXA checks traceability; the quality owner approves; ONYX publishes gaps and owners.

Managed Service Assurance Review

`managed-services.assurance-review · 6 nodes · step budget 12 · timeout 180s · 0 dispatch nodes`

ONYX frames the customer outcome; RELAY reads and assesses service assurance; HEXA verifies evidence and ownership boundaries; a service owner approves; RELAY publishes the brief.

6 of 7 cannot act at all

Only the controlled action mission contains dispatch nodes, and it has seven — six domain work items plus one RELAY coordination item, every one structurally downstream of a single human gate. Every remaining graph reads, analyses, drafts and explains. Tests assert that the Factory Flow, Working Capital, QMS and Managed Service Assurance missions contain no dispatch node.

The execution rules, in plain terms

The same engine drives all seven missions

1. **Everything ready runs together**
   The engine takes every node whose dependencies are finished and runs that whole wave at once. Eight specialists reading their evidence is one wave, not eight trips.

2. **A checkpoint after every wave**
   Progress is written down as it happens, so a run can be explained afterwards and resumed rather than restarted.

3. **Retries are bounded**
   A node may be given a second attempt, never more than five, and the limit is validated when the graph is registered rather than at runtime.

4. **Failure stops the mission**
   A failed node, a timeout, or a state where nothing can progress ends the run and says so. It does not invent a result to keep things moving.

5. **Interrupted work is recovered honestly**
   If the process dies mid-node, that node is returned to pending on resume and re-run — with its attempt count preserved, so recovery cannot be used to escape the retry budget.

6. **Approval pauses everything**
   An approval node parks the run. The decision belongs to that exact node in that exact mission and cannot be reused elsewhere.

7. **Effects need an approved ancestor**
   Before a side-effecting node runs, the engine walks the graph upward looking for an approval that actually returned yes. No ancestor, no execution — checked by the engine, not by the tool.

09 · TRUST MODEL

Why the person stays in control

Five boundaries, on every mission, with no exceptions

1 · Identity	A verified token decides the tenant and the actor
2 · Authority	The agent's allow-list AND the person's own permission
3 · Action	A side effect needs an approved ancestor node
4 · Business rules	The normal domain service still decides what is legal
5 · Evidence	Runs, nodes, events, checkpoints and audit all persist

Failure is visible

A failed node, a timeout or a deadlock is recorded and the run stops. Nothing is smoothed over to keep a demo comfortable.

Recovery is real

Checkpoints let an interrupted run continue without pretending the interrupted step had finished.

AI is fenced separately

The governed registry has 9 routable entries: eight canonical features plus the read-only Copilot still marked `in_eval`. A separate non-routable Integrations null declaration makes the absence of Integration AI explicit. A kill switch, budgets and evaluation gates constrain promotion.

Code decides, people approve

Tax, stock, quality verdicts, payroll and accounting are arithmetic. A model may phrase an explanation; it never sets the number.

Where the guarantees are actually kept

Continuous integration re-proves the tenant fence on every build, a live two-tenant probe tries to read across the boundary and fails, and all 147 permissions are reconciled against the routes that demand them. These are gates, not intentions.

How the shareable build is packaged

Railway-ready five-service topology; no live cloud URL is claimed

Public browser → Next.js web → NestJS API → PostgreSQL → Valkey

Web service	Dynamic Railway port · routes API calls to the public API URL
API service	Waits for PostgreSQL · migrates · seeds/checkpoints once · starts HTTP
Worker service	Consumes the transactional outbox independently from the API
PostgreSQL	Custom demo image with extensions, application role and all durable records
Valkey	Redis-compatible cache/coordination service, not the source of business truth

Public-demo identity

The hosted demo opens without the former sign-in screen. An explicit public-demo mode maps a small, fixed persona list to seeded tenant permissions; normal bearer-token checks remain the default outside that mode.

Repeatable boot

Deployment scripts wait for dependencies, apply all migrations, seed only when the checkpoint is absent, and expose liveness/readiness endpoints. Restarts do not blindly rebuild the database.

Current status

The repository contains Railway Dockerfiles, service descriptors, a Compose-equivalent topology and a deployment runbook. A production domain or confirmed public Railway URL is not present in the working directory, so the honest state is “deployment-ready demo packaging”, not “currently hosted”.

11 · THE HONEST BOUNDARY

What this build does and does not do

Say the right-hand column out loud before anyone finds it

Live in the repository

- Nine agents with fixed allow-lists
- 19 registered capabilities, one of them effectful
- 7 versioned, content-hashed mission graphs
- Parallel waves, checkpoints, bounded retries, recovery
- Tenant fencing and permissions, proven by CI
- Approval gates with an append-only trail
- Idempotent, approval-bound work-item dispatch
- Hash-chained audit with a real verifier

Not claimed

- No external model API is active — reasoning is deterministic
- No external connector is live; integration is simulate-driven
- No general SQL tool, and no way to create one at runtime
- No agent can widen the asking person's permissions
- No automatic customer message, payment or posting
- A dispatched work item is never described as completed work
- No quotation, lead or engineering change control exists
- The purchase cycle stops at the goods receipt

The one-line disclosure

Orchestration, ERP reads, approval gates and governed dispatch are live. Language reasoning is deterministic, and no external model API or connector is active.

Where to read further

The master stays short; each agent has its own handbook

ONYX · Supervisor
ONYX Copilot, Agent OS, AI Operations

HEXA · Governance
Organisation & master data, Administration, Integration

MICA · Sales & Product Care
Sales, Customer Care & Warranty

SPAR · Supply
Purchase, Inventory

AXLE · Planning
Engineering, Planning

KILN · Operations & Quality
Production, Quality & QMS, Maintenance

RASP · Finance & People
Accounts, Employee spend, People, Working capital

RELAY · Managed Services
Managed-service design, Transition, Service operation, Service improvement

ACHILES · Platform Assurance
Private platform status, Hourly deterministic checks, Availability evidence history

Each guide follows the same twelve pages: purpose, two module deep dives, the end-to-end business pipeline, enforced rules, runtime contract, exact tools, mission participation, a worked example, honest limits and a quick reference.

Shared glossary

Agent	A named specialist with a fixed, allow-listed tool surface.
Capability	One registered operation with a mode, an owner and a required permission.
Mission graph	A versioned recipe for steps, parallel work, joins and approval gates.
Checkpoint	A durable progress snapshot used for explanation and recovery.
Governed work item	An approved, append-only assignment — not the business change itself.
Verified value	An observed result with its method recorded, kept distinct from an estimate.

Sources: the capability registry, graph catalogue and run engine; the module services and migrations; the managed-service blueprint; automated tests; ISO/IEC 20000-1, ITIL practice guidance, NIST SP 800-61 Rev. 3, Google SRE SLO guidance and OpenTelemetry. The Cisco CoSol HTML supplied by the user informed the managed-service presentation pattern, not XELOR's implementation claims. Snapshot: 8 August 2026.